

FastAPI WebSocket Disconnect SOP

Objective

Stabilize FastAPI WebSocket connections when clients disconnect frequently due to timeout, proxy, heartbeat, or connection handling issues.

Scope

Applicable to FastAPI-based services, backend APIs, API gateways, DevOps teams, QA teams, and platform teams supporting production, staging, or synthetic incident workflows.

Pre-requisites

1. Access to FastAPI application logs, deployment logs, and API gateway or ingress logs.
2. Access to monitoring dashboards such as Grafana, Prometheus, APM, or cloud monitoring.
3. Permission to view application configuration, environment variables, and deployment status.
4. API testing tool such as curl, Postman, Swagger UI, or automated smoke test scripts.
5. Escalation contact for backend, DevOps, database, and security teams.

Incident Metadata

Field	Value
Ticket ID	T042
Issue	FastAPI websocket disconnect issue
Category	API
Service	FastAPI
Severity	Medium
Status	In Progress
Description	Websocket disconnects frequently
Expected Resolution	Check connection handling logic

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L020	5/1/2026 10:10	ERROR	Websocket disconnected	API_WS
L040	5/1/2026 10:20	ERROR	Websocket disconnected	API_WS

Procedure

1. Confirm disconnect frequency, client type, route, close code, and incident window.
2. Review application logs for WebSocket disconnect events and exception details.
3. Validate client heartbeat, ping/pong behavior, and reconnect logic.
4. Check ingress, load balancer, and proxy idle timeout settings.
5. Confirm application handles WebSocketDisconnect exceptions cleanly.
6. Review worker count, sticky session requirements, and scaling behavior.
7. Tune heartbeat or proxy timeouts and fix connection handling logic.
8. Monitor active connections, disconnect count, and reconnect success rate.

Common Commands

```
curl -i -X GET http://<host>/health
curl -i -X POST http://<host>/<endpoint> -H "Content-Type: application/json" -d @payload.json
kubectl get pods -n <namespace> | grep fastapi
kubectl logs <fastapi-pod> -n <namespace> --tail=100
kubectl describe pod <fastapi-pod> -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Frequent disconnects	Idle timeout	Tune proxy and heartbeat settings
Disconnect on scale out	No sticky session or shared state	Use shared state or sticky routing
Server exception	Unhandled WebSocketDisconnect	Handle disconnect path cleanly
Client reconnect loop	Bad client retry logic	Fix reconnect backoff

Best Practices

1. Track WebSocket connection metrics.
2. Handle disconnect exceptions without noisy errors.
3. Align proxy timeout with heartbeat interval.
4. Use shared state for multi-replica WebSocket services.

References

[FastAPI Docs](#) | [Internal Logs CSV](#) | [Tickets CSV](#) | [Monitoring Dashboards](#) | [API Gateway Logs](#) | [Backend Runbook](#)